

Assignment 2 – Programming Concepts – Lotto Game

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Introduction

This is a program about a Lotto game. The program allows the user to enter 6 unique numbers between 1 to 40 and then the numbers will be compared to the randomly generated 6 unique winning numbers.

The program will display the user's numbers.

The program will display 6 unique randomly generated numbers and also display the numbers that matched.

Data types

Integer

Boolean

String

Arrays

playernumber

randomnum

usernumbers

LotteryNumbers

Code for the program

```
package Assignments;

import javax.swing.JOptionPane;

import javax.swing.JTextArea;

/**
 *
 * @author NIRVA
 */

public class LottoGame {

    /**
     * @param args the command line arguments
     */

    public static void main(String[] args) {

        //main program

        //global

        JOptionPane.showMessageDialog(null, "Welcome to Lotto Game\n Lottotech wishes
you good luck");

        JOptionPane.showMessageDialog(null,"You must enter 6 unique numebrs");

        //initializing for the display of the methods

        int usernumbers[] = inputplayernumber();

        int LotteryNumbers[] = winningnumbers();

        int numberMAatching = MatchingNumbers(usernumbers, LotteryNumbers);

        int reward;

        Results(usernumbers, LotteryNumbers, numberMAatching);
```

```

//
reward = WinnerReward(numberMatching);
if(reward > 0){
    JOptionPane.showMessageDialog(null, "Congrats, you won RS: " + reward);
}else{
    JOptionPane.showMessageDialog(null, "you didnt win anything");
}

System.exit(0);
}

//public be able to return a value to the main
//method for the player to display their number
public static int[] inputplayernumber() {
    String input;
    int counter1;
    int[] playernumber = new int [6];
    int output;
    boolean valid;
    boolean duplicate;
    int counter4;
    //to enter player numbers

    JTextArea dongdong = new JTextArea();
    String output3 = "Array cells" + "\t" + "User input" + "\n";
    //control structures

```

```

for(counter1 = 0; counter1 < 6; ){
    valid = false;
    while (!valid) {
        input = JOptionPane.showInputDialog("enter number between 1 and 40" + " : " +
(counter1 + 1));
        output = Integer.parseInt(input);

        if (output >= 1 && output <= 40 ) { //when using || command it is registering the
large numbers...ask teachers why
            duplicate = false;
            for (counter4 = 0; counter4 < counter1; counter4++) {
                if (playernumber[counter4] == output){
                    duplicate = true;
                    break;
                }
            }
            if (!duplicate){
                playernumber[counter1] = output;

                valid = true;
            }else {
                JOptionPane.showMessageDialog(null, "Lottotech machine is not allowed to
generate the same number! \nGenerating a different number in range 1- 40\nEnter a unique
number!");
                counter1=(counter1 - 1);
            }
        }else{

```

```
        JOptionPane.showMessageDialog(null,"number: " + output + " is out of range  
enter again, enter number between 1-40");
```

```
    }
```

```
    output3+= "\n number [" + (counter1+1) + "]" + "\t" + output;
```

```
    counter1++;
```

```
}
```

```
dongdong.setText(output3);
```

```
JOptionPane.showMessageDialog(null, dongdong);
```

```
}
```

```
return playernumber;
```

```
}
```

```
//method for generating random numbers
```

```
public static int[] winningnumbers(){
```

```
    int randomnum[] = new int [6];
```

```
    int counter2,counter5;
```

```
    int randomGen;
```

```
    boolean valid, duplicate;
```

```
    JTextArea dingding = new JTextArea();
```

```
    String output2 = "Array cells" + "\t" + "Winning numbers" + "\n";
```

```
    for( counter2 = 0; counter2 < randomnum.length; counter2++ ){
```

```
        valid = false;
```

```
        while(!valid){
```

```

        randomGen = (int)(Math.random()*40) + 1;
        duplicate = false;
        for(counter5 = 0; counter5 < counter2; counter5++){
            if(randomnum[counter5] == randomGen){
                duplicate = true;
                break;
            }
        }
        if(!duplicate) {
            randomnum[counter2] = randomGen;
            valid = true;
        }

        output2 = output2 + "number [" + (counter2 + 1) + "]" + "\t" + randomnum[counter2]
+ "\n";

    }

    dingding.setText(output2);

    JOptionPane.showMessageDialog(null, dingding);

}

return randomnum;
}

```

[//method to see if the users and the lotto numbers match](#)

```

public static int MatchingNumbers(int[] usernumbers, int[] LotteryNumbers) {

```

```

int numberMAatching = 0;

int counter6,counter7;

JOptionPane.showMessageDialog(null,"Now the machine will give 6 winning numbers");

for(counter6 = 0; counter6 < 6; counter6++){
    for(counter7 = 0; counter7 < 6; counter7++){
        if(usernumbers[counter6] == LotteryNumbers[counter7]){
            numberMAatching++;
            break;
        }
    }
}

return numberMAatching;
}

```

//method to display the numbers of the user numbers, lotto numbers, and the numbers that matched

```

public static void Results(int[] usernumbers, int[] LotteryNumbers, int numberMAatching) {

    String UserNumberInput = "";
    String WinningNumbersLottery = "";
    int counter8;

    JTextArea bing = new JTextArea();

    String output11 = "Array cells" + "\t" + "Winning numbers" + "\n";

```

```

for(counter8 = 0; counter8 < 6; counter8++) {
    UserNumberInput += usernumbers[counter8] + " ";
}
for(counter8 = 0; counter8 < 6; counter8++) {
    WinningNumbersLottery += LotteryNumbers[counter8] + " ";

}

output11 = "User NUmbers: " + UserNumberInput + "\t" + "Winning Lottery Numbers: "
+ WinningNumbersLottery + "\t" + "Lucky numbers: " + numberMAatching;

bing.setText(output11);

JOptionPane.showMessageDialog(null, bing);
}

```

//rewards for the numbers that match the winning numbers.

```

public static int WinnerReward(int numberMAatching){
    switch (numberMAatching){

        case 3:
            return 100;

        case 5:
            return 1000;

        case 6:
            return 1000000;
    }
}

```

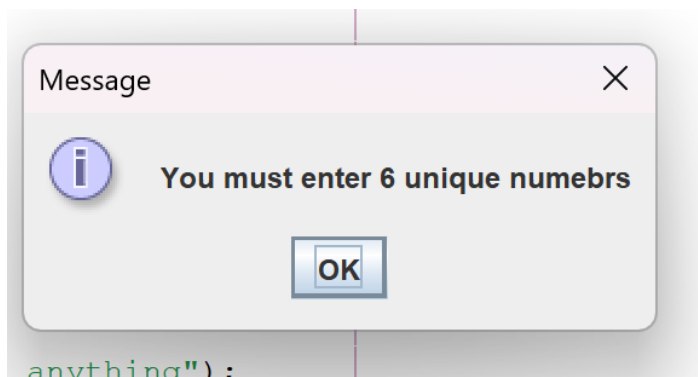
```
        default:
            return 0;
    }
}
}
```

Screenshots

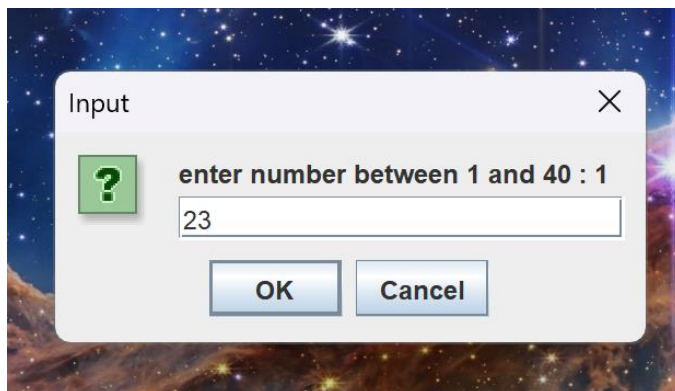
Welcome screen



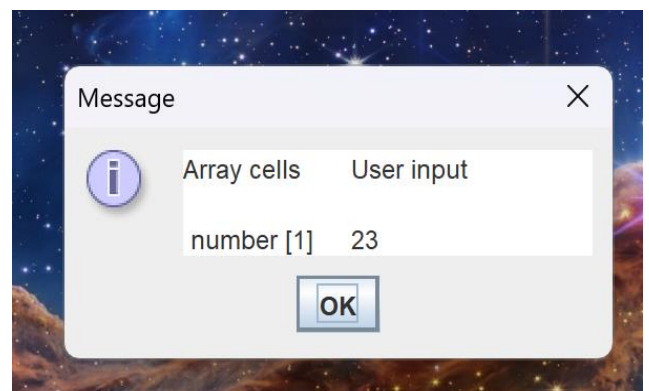
Message



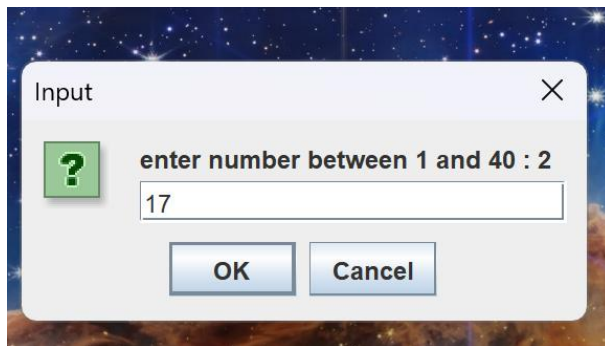
User 1st the numbers



Number is stored in the array.

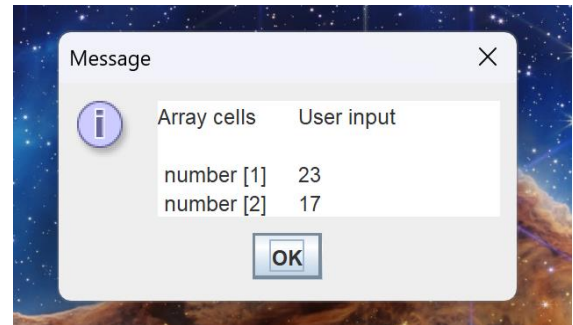


Entering 2nd number



An input dialog box titled "Input" with a close button (X) in the top right corner. It contains a green square icon with a white question mark. The text "enter number between 1 and 40 : 2" is displayed above a text input field. The number "17" is entered in the field. Below the field are two buttons: "OK" and "Cancel".

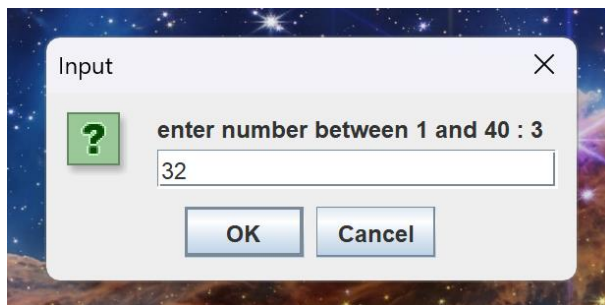
Number is stored in the array.



A message dialog box titled "Message" with a close button (X) in the top right corner. It contains an information icon (i) and a table with two columns: "Array cells" and "User input". The table has two rows: "number [1]" with value "23" and "number [2]" with value "17". Below the table is an "OK" button.

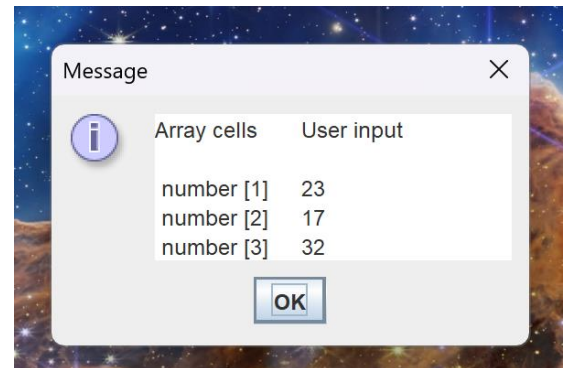
Array cells	User input
number [1]	23
number [2]	17

Entering 3rd number



An input dialog box titled "Input" with a close button (X) in the top right corner. It contains a green square icon with a white question mark. The text "enter number between 1 and 40 : 3" is displayed above a text input field. The number "32" is entered in the field. Below the field are two buttons: "OK" and "Cancel".

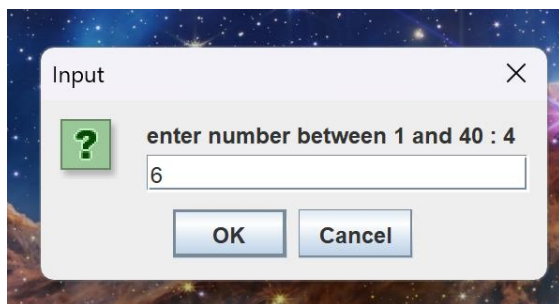
Number is stored in the array.



A message dialog box titled "Message" with a close button (X) in the top right corner. It contains an information icon (i) and a table with two columns: "Array cells" and "User input". The table has three rows: "number [1]" with value "23", "number [2]" with value "17", and "number [3]" with value "32". Below the table is an "OK" button.

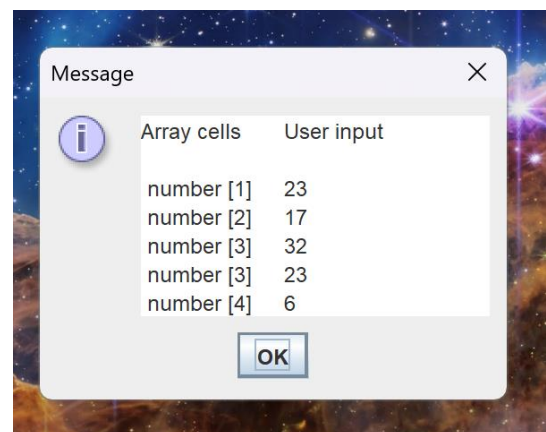
Array cells	User input
number [1]	23
number [2]	17
number [3]	32

Entering 4th number.



An input dialog box titled "Input" with a close button (X) in the top right corner. It contains a green square icon with a white question mark. The text "enter number between 1 and 40 : 4" is displayed above a text input field. The number "6" is entered in the field. Below the field are two buttons: "OK" and "Cancel".

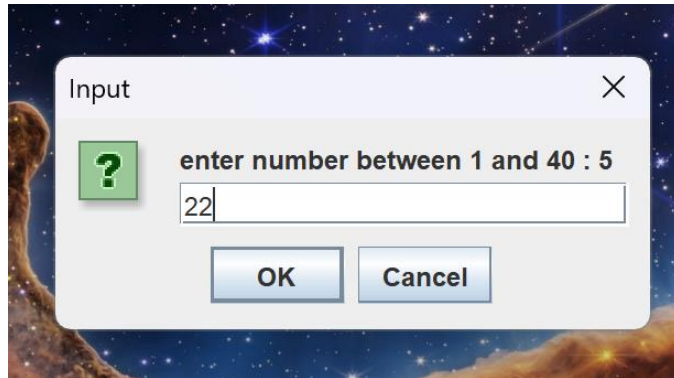
Number is stored in the array.



A message dialog box titled "Message" with a close button (X) in the top right corner. It contains an information icon (i) and a table with two columns: "Array cells" and "User input". The table has four rows: "number [1]" with value "23", "number [2]" with value "17", "number [3]" with value "32", and "number [4]" with value "6". Below the table is an "OK" button.

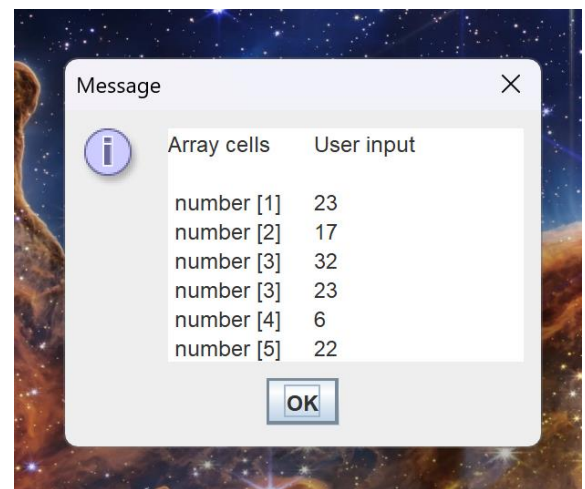
Array cells	User input
number [1]	23
number [2]	17
number [3]	32
number [4]	6

Entering the 5th number.



An input dialog box titled "Input" with a close button (X) in the top right corner. It features a green square icon with a white question mark on the left. The text "enter number between 1 and 40 : 5" is displayed. Below the text is a text input field containing the number "22". At the bottom are two buttons: "OK" and "Cancel".

Number is stored in the array.

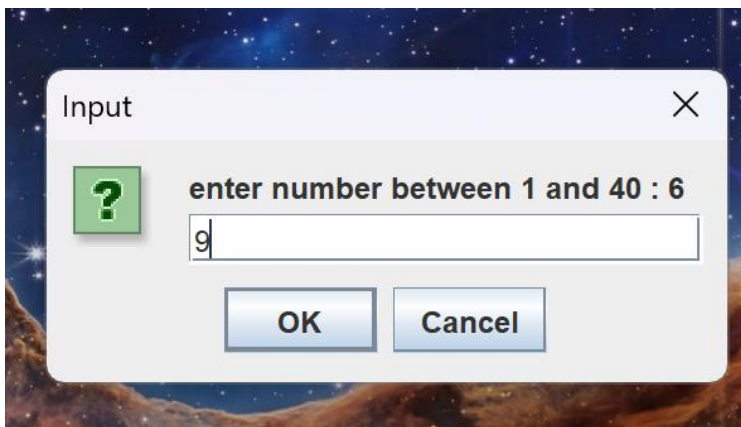


A message dialog box titled "Message" with a close button (X) in the top right corner. It features a blue circular icon with a white 'i' on the left. The dialog contains a table with two columns: "Array cells" and "User input".

Array cells	User input
number [1]	23
number [2]	17
number [3]	32
number [3]	23
number [4]	6
number [5]	22

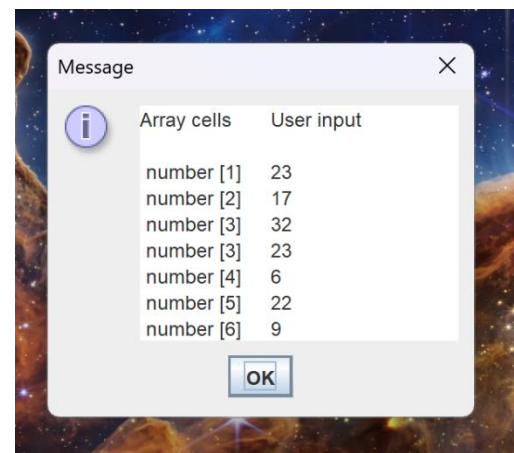
At the bottom of the dialog is an "OK" button.

Input for 6th number



An input dialog box titled "Input" with a close button (X) in the top right corner. It features a green square icon with a white question mark on the left. The text "enter number between 1 and 40 : 6" is displayed. Below the text is a text input field containing the number "9". At the bottom are two buttons: "OK" and "Cancel".

Number is stored in the array.

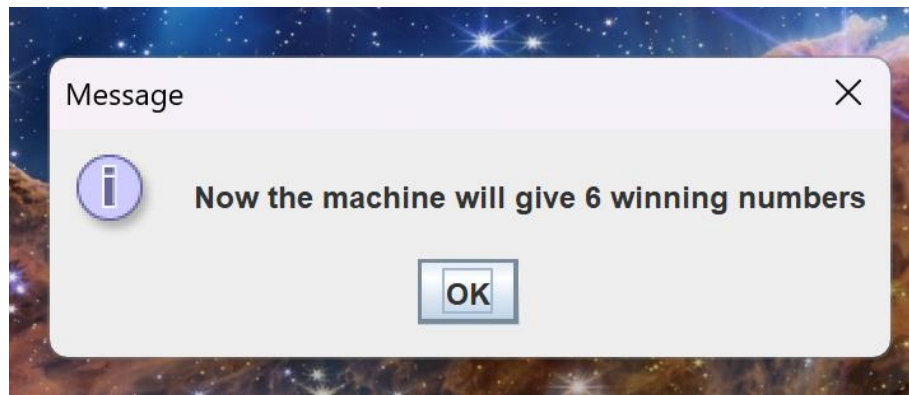


A message dialog box titled "Message" with a close button (X) in the top right corner. It features a blue circular icon with a white 'i' on the left. The dialog contains a table with two columns: "Array cells" and "User input".

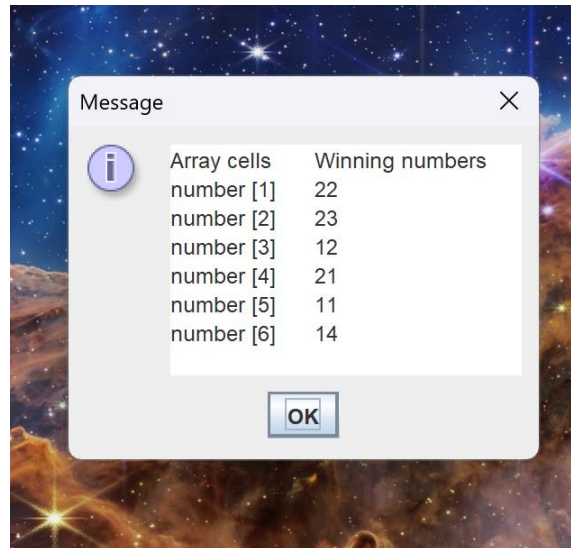
Array cells	User input
number [1]	23
number [2]	17
number [3]	32
number [3]	23
number [4]	6
number [5]	22
number [6]	9

At the bottom of the dialog is an "OK" button.

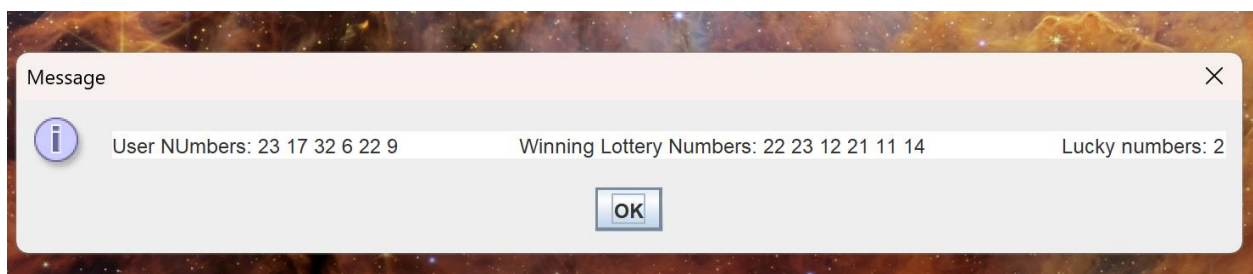
Message after entering all the 6 numbers



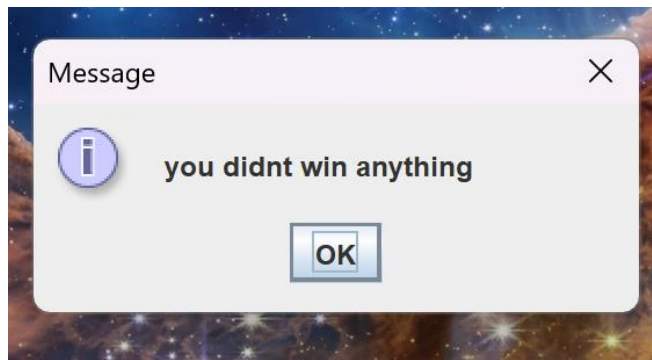
Displaying the winning numbers.



The program will display the numbers entered by the user, the winning numbers, and the numbers of the users that match with the Lotto game winning numbers.



If no numbers match, this message will display



Duplicate

If the number is entered twice, this message will appear.

